



Distribution of indium and gallium in sphalerite from skarn and hydrothermal vein deposits in the Hwanggangri mineralized district, South Korea

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ABSTRACT

The geochemistry of sphalerite from seven metallic ore deposits in the Hwanggangri mineralized district in South Korea was studied, with a special focus on the concentrations of In and Ga. The chemical compositions were analyzed using laser ablation-inductively coupled plasma-mass spectroscopy and compared based on deposit type (skarn and hydrothermal vein) and ore species (W, Mo, Zn–Pb, and W–Zn–Pb). In hydrothermal W and W–Zn–Pb vein deposits, In was highly concentrated, with averages of up to 696 ppm, whereas Mo and Zn–Pb skarn deposits produced low concentrations below 43 ppm on average. In contrast, the Ga concentrations were quite low throughout the deposits and did not exceed 16 ppm on average. Indium showed a clear positive correlation with Cu and was strongly associated with the development of chalcopyrite-disease, with Ag and Sn potentially playing subordinate roles in the substitution. The binary correlations between trace elements suggested that principal coupled substitutions, such as $3\text{Zn}^{2+} \leftrightarrow 2(\text{Cu}, \text{Ag})^+ + \text{Sn}^{4+}$ and $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{In}^{3+}$, prevailed in In-bearing sphalerite. A discernable weak positive correlation is observed between In and Cd, mostly in the W-bearing deposits, implying that Cd, a common constituent in sphalerite, could be used as an exploration proxy in some cases. The general tendency of high In and low Ga concentrations in the sphalerites from the Hwanggangri mineralized district indicates relatively high temperature of mineralizing fluids, generally above 250 °C, which favored the prevalence of In-chloride complexes over Ga-hydroxide complexes. Additionally, the In enrichment in the hydrothermal vein ores rather than the skarn ores emphasizes the importance of the direct contribution of magmatic fluids to ore mineralization under high temperature conditions with little influence from the host rock.

1. Introduction

Today, minerals containing critical elements, such as Li, V, Co, In, Nb, Re, Se, Te, Sb, Be, F, Ga, Ge, rare earth elements, and platinum-group elements, are becoming increasingly important in high-tech industries, as these elements are used in semiconductors, optical fibers, LED flat panels, superconducting materials, solar cells, special alloys, lenses, and batteries (Bobba et al., 2020). Consequently, the importance of securing these resources is increasing because of their low concentrations in the crust and the potential instability of the global supply chain (Fortier et al., 2020). Among these elements, the economic value of In and Ga, produced from sphalerite, is increasing from an industrial

perspective (USGS, 2020). Consequently, many studies have been conducted on the occurrence characteristics of these elements in sphalerite.

It is known that there are approximately 10 minerals containing In as a main element, including roquesite (CuInS_2) and laforetite (AgInS_2). However, these minerals are produced in very small quantities and are not easily concentrated into ore deposits. Most In is produced as a by-product of Cu, Sn, and Zn, with sphalerite, chalcopyrite, cassiterite, and stannite, containing approximately 1–100 ppm (Alfantazi and Moskalyk, 2003; Werner et al., 2017). The major In-related deposit types include polymetallic epithermal, volcanogenic massive sulfide, proximal skarn, granite-related, porphyry, and sediment-hosted stratiform Cu deposits (Cook et al., 2009; Jovic et al., 2011; Ye et al., 2011; Werner

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et al., 2017; Carvalho et al., 2018). Many polymetallic deposits contain significant amounts of In, which is concentrated in the sphalerite of Sn-rich deposits from several countries, including Japan (the Toyoha, Ashio, and Akenobe deposits), China (the Dulong and Dachang deposits), and Bolivia (the Potosi, Huari Huari, and Bolivar deposits) (Murakami and Ishihara, 2013). Carvalho et al. (2018) reported that In is more abundantly present in Cu-enriched ores compared to Zn-enriched ores within volcanogenic massive sulfide deposits. Moreover, a strong correlation is identified between the concentrations of these two elements. Ishihara et al. (2006) explained that, in Japan, In mineralization is closely related to skarn and vein deposits influenced by acidic igneous bodies. They proposed that deep igneous activity that penetrated the accretionary bodies in subduction zone environments played an important role.

Minerals containing Ga as their main element, such as gallite (CuGaS_2), galloboeudantite ($\text{PbGa}_3(\text{AsO}_4)(\text{SO}_4)(\text{OH})_6$), sohngeite ($\text{Ga}(\text{OH})_3$), and tsumgallite ($\text{GaO}(\text{OH})$) are extremely rare, and these minerals do also not form ore bodies. Gallium typically exists as a trace element in various minerals. It is mainly found in Al-hydroxide minerals, such as boehmite, diaspore, and gibbsite, which constitute bauxite (Dutrizac et al., 1986; Bernstein and Waychunas, 1987; Moskalyk, 2003). These minerals currently account for approximately 90 % of global Ga production (Rongguo et al., 2016). Gallium is also produced in significant amounts as a by-product of deposits with large quantities of sphalerite (Moskalyk, 2003). In addition, Ga is enriched in bornite, enargite, galena, and luzonite (Burton and Culkin, 1978) and is produced by replacing the Fe in chalcopyrite (Foley et al., 2017). Significant

amounts of Ga, along with Li and rare earth element (REE), are enriched in the carbonaceous materials composed of Al-enriched kaolinite, boehmite, and chlorite group minerals in the Jungar Coal field in China (Sun et al., 2016). Given the association of In and Ga as by-products of sphalerite, it is important to re-evaluate the potential for these elements in sphalerite deposits that were previously developed only for Zn. Determining the types of deposits or ore species that are enriched with In and Ga is critical for exploration and development efficiency.

The Hwanggangri mineralized district (HMD), a representative mineralized district in South Korea, hosts numerous deposits of skarn and hydrothermal vein types, accompanied by various ore species such as W, Mo, Zn-Pb, Au-Ag, fluorite, talc, and polymetallic deposits (Reedman et al., 1973; Yun et al., 1986). However, there have been few studies on the abundances of critical elements in this district. To the best of our knowledge, the only study to date was by Heo et al. (2008), who conducted a preliminary study on the concentrations of In, Re, Ga, Ge, Se, Te, Y, Eu, and Sm in ores from several deposits in the district.

This study analyzes the major and minor/trace elements of sphalerite from seven deposits in the HMD using laser ablation-inductively coupled plasma-mass spectrometry (LA-ICP-MS). The results were used to compare the concentrations of In and Ga in sphalerite based on the deposit type (skarn or hydrothermal vein) and ore species (W, Mo, Zn-Pb, or W-Zn-Pb), address the substitution mechanisms of their mineralization processes, and identify relevant and useful exploration proxies for In and Ga mineralization in sphalerite.

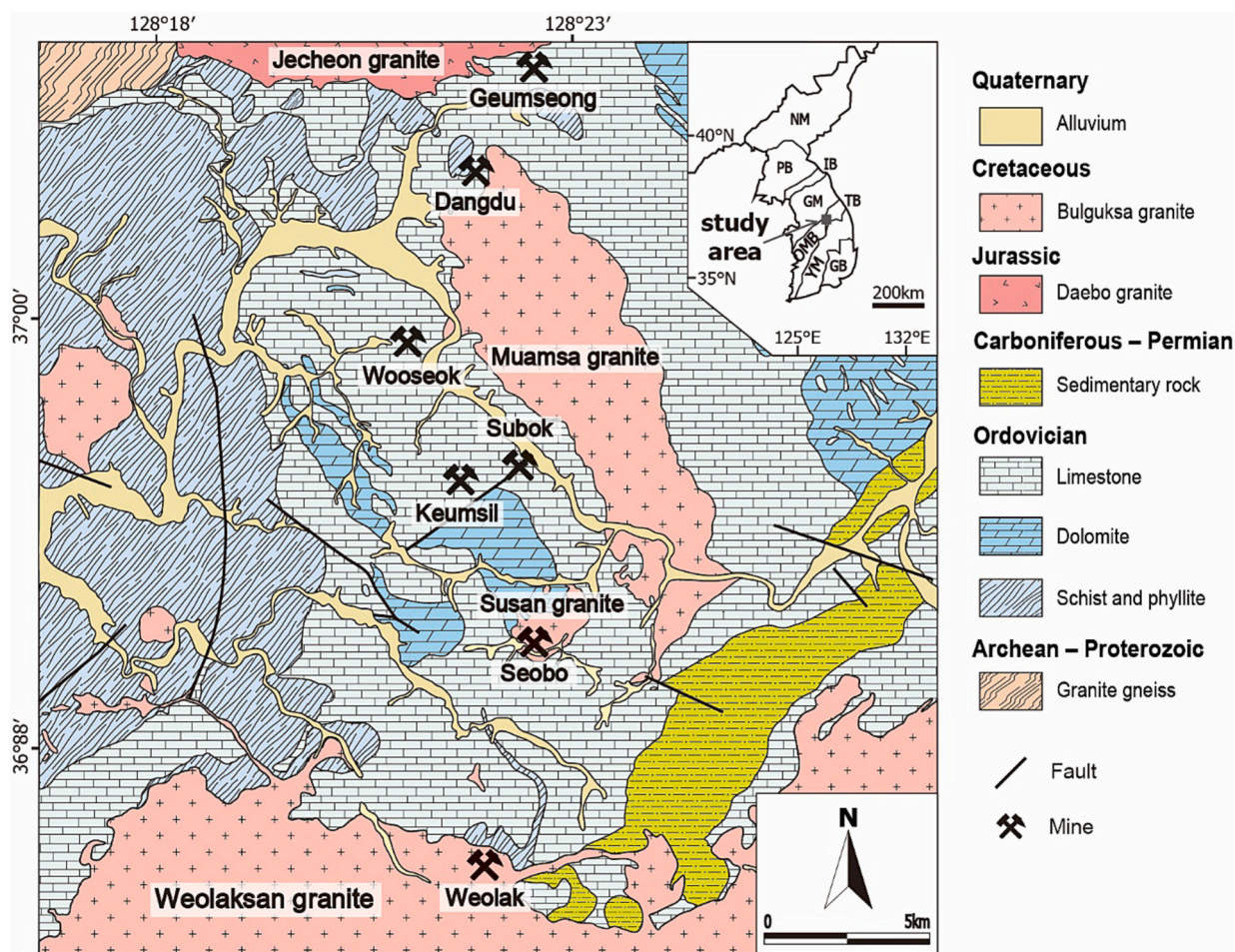


Fig. 1. Regional geologic map showing the distribution of studied ore deposits in the Hwanggangri mineralized district (modified from Im et al., 2018). NM: Nangnim Massif, PB: Pyeongnam Basin, IB: Imjingang Belt, GM: Gyeonggi Massif, TB: Taebaeksan Basin, OMB: Okcheon Metamorphic Belt, YM: Yeongnam Massif, GB: Gyeongsang Basin.

2. Geological background

The geology of the southern Korean Peninsula can be classified into four regions running from northwest to southeast: the Gyeonggi Massif, Okcheon Belt, Yeongnam Massif, and Gyeongsang Basin (Fig. 1). The Gyeonggi and Yeongnam massifs are Precambrian basements of metamorphic complexes composed primarily of schist and gneiss. These metamorphic rocks are divided into two groups: Archean–Paleoproterozoic basement rocks and Neoproterozoic supracrustal rocks, which underwent metamorphism during the Paleoproterozoic era and the Permian–Triassic period, respectively (Cluzel et al., 1990; Chough et al., 2000; Sagong et al., 2003, 2005).

The Okcheon Belt is a fold-thrust belt that runs over 350 km in a NE–SW direction and is approximately 50 km wide. This belt marks the boundary between the Gyeonggi and Yeongnam massifs and was formed during the Early Triassic by the collision of the Sino-Korean Craton (or North China Block) and the South China Block (Cluzel et al., 1990; Koh and Kim, 1995; Chough et al., 2000; Ree et al., 2001). The Okcheon Belt is subdivided into two regions based on petrological criteria, such as lithology and metamorphic grade: the Okcheon metamorphic belt in the southwest and the Taebaeksan Basin in the northeast (Chough et al., 2000; Suzuki et al., 2006; Choi et al., 2012). The Okcheon metamorphic belt mainly comprises low-to-middle grade metasedimentary and metavolcanic rocks of the Okcheon Supergroup (Oh et al., 2004; Suzuki et al., 2006). In the Taebaeksan Basin, the Joseon Supergroup of the Lower Paleozoic (primarily limestone) and the Pyeongan Supergroup of the Upper Paleozoic (mainly shale and sandstone) are distributed over a wide area. They show an unconformity relationship with the Yeongnam Massif. The Gyeongsang Basin, located in the southeastern part of the Korean Peninsula, comprises Cretaceous alluvial, lake sedimentary, and acidic volcanic rocks from the Cretaceous–Tertiary period (Chang et al., 1997; Chough et al., 2000).

The Hwanggangri mineralized zone (HMZ) is located in the Okcheon Belt and comprises Precambrian gneiss from the Gyeonggi Massif in the basement and sedimentary or metasedimentary rocks from the Okcheon and Chosun supergroups. The zone is intruded by Jurassic Daebok and Cretaceous Bulguksa granites. The Okcheon Supergroup mainly consists of phyllite, slate, schist, and quartzite, with minor intercalations of fine-grained crystalline limestone, and often exhibits microfold structures (Chough et al., 2000; Suzuki et al., 2006; Kang et al., 2012). The Samtaesan Formation, composed of calcitic marble, dolomitic marble, slate, argillaceous hornfels, amphibolite, and limestone, forms the upper part of the Chosun Supergroup. This is where most of the deposits are hosted (Park and Park, 1979). The gray or dark gray calcitic and dolomitic marbles have undergone moderate to high degrees of recrystallization, resulting in the formation of silicate minerals such as tremolite, diopside, talc, wollastonite, olivine, phlogopite, and serpentine (Shin and Lee, 2002).

The mineralized district is closely associated with igneous bodies, such as the Cretaceous Muamsa, Susan, and Weolaksan granites, which are widely distributed around it. The Muamsa granite intrudes the Joseon Supergroup in a NNW–SSE direction and is composed of medium-to-coarse grained biotite granite and porphyritic granite (So and Yun, 1992). Although it appears to be a stock on the surface, gravity exploration confirmed that the granite body intruded a wider area than previously thought (Shin et al., 2014). In the outer part of the igneous body, fine-grained granite with myrmekitic and perthite textures developed, and miarolitic cavities can be observed where the fluid separated and escaped during rapid magma cooling (So and Yun, 1992; Lee et al., 2010). The Susan granite is located 1 km southwest of the Muamsa granite, and penetrates the Samtaesan Formation as a small-scale cupola. It has a more homogeneous texture and composition than other granites, and porphyries are rarely observed; however miarolitic cavities have been observed at the periphery of the igneous bodies (So and Yun, 1992, 1994). The Weolaksan granite is located south of the HMZ and forms a large batholith comprising biotite and

biotite–hornblende granite. The granite shows a medium-to-coarse grained lithology, and its porphyritic texture develops toward its periphery (So and Yun, 1992; Lee et al., 2010).

Although these three granites differ in shape and distribution, they share similar lithologies and compositions. They are all coarse-grained biotite granites, which are mainly composed of quartz, alkali feldspar, plagioclase, and biotite. The reported ages of the granite bodies fall within the narrow range of 88–90 Ma, in the Late Cretaceous (So and Yun, 1992; Lee et al., 2010).

3. Ore deposits

The HMZ hosts various metallic and non-metallic deposits, including W, Mo, Zn–Pb, Au–Ag, W–Zn–Pb, fluorite, and talc deposits. They are primarily grouped into skarn and hydrothermal vein deposits (Reedman et al., 1973; Choi et al., 2019). Seven ore deposits ranging from north to south of the HMZ, namely the Geumseong, Dangdu, Wooseok, Subok, Keumgil, Seobo, and Weolaksan deposits were studied (Fig. 1). They formed under similar geological conditions but represent different ore species and deposit types. Table 1 summarizes the general characteristics of these deposits and the analyzed samples.

3.1. Seobo deposit

The Seobo deposit is a fissure-filling hydrothermal W deposit hosted by the Susan granite (Fig. 1). Greisenization developed, establishing mineral assemblages consisting mainly of quartz and muscovite, with scheelite, wolframite, and molybdenite. The paragenetic sequence of the ore minerals is characterized by an early W–Mo stage and a late base metal sulfide stage (KORES (Korea Resources Corporation), 1981; So and Yun, 1994). Scheelite occurs at the contact between the host rock and quartz veins, and sulfide minerals such as pyrite, arsenopyrite, sphalerite, and galena occur in the veins (Fig. 2a, b, g, h). The pyrite and arsenopyrite exhibit anhedral to subhedral shapes, and sphalerite was formed later by replacing them or filling gaps within the cracks. In most of the sphalerite, moderate chalcopyrite exsolution textures developed, forming irregular spots (Fig. 2h). The formation temperature is estimated to have ranged from 285 to 370 °C in the W–Mo stage and from 200 to 320 °C in the base metal sulfide stage (So and Yun, 1994).

3.2. Weolaksan deposit

The Weolaksan deposit is a fissure-filling W–Mo deposit located in the southern region of the HMZ and is closely associated with the Weolaksan granite (Fig. 1). Wall-rock alteration along the vein margins has produced selvages containing sericite and chlorite. Additionally, a thin greisen zone containing sericite, pyrite, quartz, and a small amount of molybdenite borders the veins near the granite (So et al., 1983). Initially, wolframite occurred alongside magnetite, then molybdenite, pyrite, arsenopyrite, and scheelite crystallized successively (Fig. 2c). Regardless of their size, most of the sphalerite contain exsolved chalcopyrite in the form of bead chains or veinlets (Fig. 2i). The homogenization temperature of the fluid inclusions in the quartz, beryl, and scheelite is 220–380 °C, and the salinity is 3.7–11.1 wt% NaCl eq. (Lee and Park, 1982; So et al., 1983).

3.3. Wooseok deposit

The Wooseok deposit, which is adjacent to the Muamsa granite (Fig. 1), exhibits a vertical pattern of skarn and hydrothermal vein ores with W–Fe and Zn–Pb–Cu mineralization, respectively (Park and Park, 1979; Im et al., 2018). Andradite garnet and clinopyroxene initially dominate the skarn, indicating an oxidizing environment. These are replaced or cut by later stage assemblages such as magnetite–hematite, molybdenite–scheelite–wolframite, and pyrite–chalcopyrite–pyrrhotite–sphalerite (Fig. 2d). An exsolution texture comprising

Table 1

Geological features of the studied ore deposits in the Hwanggangri mineralized district.

Deposit	Mode of occurrence	Main commodities	Related igneous rock	Host rock	Ore mineral assemblage	$\delta^{34}\text{S}_{\text{CDT}}$ (‰) (sulfides)	Formation temperature (°C)	Reference
Weolak	Vein	W–Mo	Weolaksan	Limestone	Sch, Mo, Bi, Py, Sph, Mc, Asp, Cp, Gn	3.0–7.6	220–380 ¹	Lee and Park, 1982; So et al., 1983
Seobo	Vein	W	Susan	Limestone	Sch, Py, Gn, Sph, Asp, Cp	2.9–6.5	285–370 ¹	So and Yun, 1994
Subok	Vein	W–Zn–Pb	Muamsa	Limestone	Sph, Gn, Sch, Wf, Po, Py, Asp, Cp	0.9–6.2	223–362 ¹	Yun et al., 1986; Im et al., 2022
Keumsil	Vein	Pb–Zn–(Au–Ag)	Muamsa	Limestone	Gn, Sph, Po, Asp, Py	5.6–9.9	255–313 ¹	So and Yun, 1992
Wooseok	Vein, Skarn	W–Fe skarn Zn–Pb–Cu vein	Muamsa	Limestone	Sph, Gn, Po, Py, Cp, Sch, Wf, Mt., Ht	5.1–6.8	267–360 ¹	Park and Park, 1979; Im et al., 2018
Geumseong	Skarn	Mo	Muamsa	Dolomite and limestone	Mo, Po, Sph, Py, Gn, Cp	n.a.	300–400 ²	Choi et al., 2007; Shin et al., 2014
Dangdu	Skarn	Zn–(Pb)	Muamsa	Dolomite and limestone	Sph, Po, Py, Mt., Gn, Cp	n.a.	<400 ²	Lim et al., 2013

n.a. = not available. sph: sphalerite, py: pyrite, asp: arsenopyrite, cp: chalcopyrite, gn: galena, po: pyrrhotite, mt: magnetite, ht.: hematite, mb: molybdenite, bi: native bismuth, sch: scheelite, wf: wolframite, mc: marcasite.

¹ Homogenization temperature of fluid inclusion.

² Mineral phase stability temperature.

chalcopyrite commonly develops in sphalerite that coexists with pyrrhotite (Fig. 2j). The tiny grains of the exsolved chalcopyrite form a slightly linear arrangement, whereas the pyrrhotite is randomly distributed in various sizes. The mineralization temperatures obtained from the homogenization temperature of the fluid inclusions are 279–360 °C for the W–Mo ores and 267–311 °C for the Zn–Pb ores (Park and Park, 1979).

3.4. Subok deposit

The Subok deposit consists of W and Zn–Pb mineralization, which develops along the fissures of carbonate rocks (Fig. 1). Quartz veins containing scheelite and small amounts of pyrite can be observed. Arsenopyrite occurs in massive form, often with pyrite and chalcopyrite (Fig. 2e). Sphalerite replaced or infilled cracks in the pyrite and arsenopyrite and was postdated by pyrrhotite and galena (Fig. 2k). Exsolution textures commonly develop in sphalerite, and chalcopyrite crystallizes along cracks in sphalerite. The homogenization temperatures of the fluid inclusion in quartz range from 237 to 347 °C with a salinity of 2.6 to 10.8 wt% NaCl eq. (Yun et al., 1986).

3.5. Keumsil deposit

The Keumsil deposit is located in the central HMZ, and is also a fissure-filling Zn–Pb–(Au–Ag) deposit (Fig. 1). It is located farther from the Muamsa granite than the other deposits. The mineralized veins, which fill northeast-trending fissures in the limestones of the Samtaesan Formation, primarily consist of a single generation of quartz and carbonates with various sulfide and sulfosalt minerals in addition to electrum and native silver (So and Yun, 1992). During the early stage, sulfide minerals such as pyrite and arsenopyrite were the predominant crystallized minerals. These minerals were later replaced by pyrrhotite, sphalerite, galena, chalcopyrite, and stannite (Fig. 2f). The sphalerite was locally crystallized with brecciated arsenopyrite and pyrite, and chalcopyrite exsolution texture is rarely observed (Fig. 2l). The temperature of early mineralization, estimated from fluid inclusions, is in the range of 310–380 °C, and it decreases sharply to 210–250 °C in the late stage. The calculated salinity of the fluid inclusions is in the range of 1.2–7.2 wt% NaCl eq. (So and Yun, 1992).

3.6. Geumseong deposit

The Geumseong deposit in the north of the HMZ (Fig. 1) is a skarn Mo deposit. Recent investigations based on mineral assemblages,

geochemical characteristics, age dating, and gravitational exploration have revealed that the associated igneous rock of the deposit is the Cretaceous Muamsa granite rather than the geographically closer Jurassic Jecheon granite (Choi et al., 2007; Shin et al., 2014). In the early stages, skarnization resulted in the formation of garnet, pyroxene, and epidote, which were later replaced by sulfide minerals (Fig. 3a). The molybdenite formed a disseminated texture, accompanied by a small amount of magnetite (Fig. 3b). Pyrite crystallized with sphalerite, and was later replaced by pyrrhotite and chalcopyrite (Fig. 3d–f). The exsolution texture of the chalcopyrite in the sphalerite is insignificant.

The associated igneous rocks of the Geumseong deposit are highly fractionated (> 73 wt% SiO₂), and early skarnization occurred in a relatively strong oxidizing environment (Choi et al., 2007). Based on the coexisting mineral assemblages and thermodynamic data, skarnization is estimated to have occurred at 400–500 °C, while sulfide mineralization would have occurred at lower temperatures of 300–400 °C (Choi et al., 2007).

3.7. Dangdu deposit

The Dangdu deposit is a Zn–Pb skarn deposit that developed in the Ordovician limestone of the Samtaesan Formation and is located in the north of the HMZ (KORES (Korea Resources Corporation), 1979; Lim et al., 2013; Jeong et al., 2014) (Fig. 1). Mineralization is considered to have occurred in a single event that can be separated into an early skarn stage and a late ore mineralization stage (Fig. 3c). Initially, skarn minerals such as pyroxene, garnet, epidote, chlorite, and tremolite formed, followed by ore minerals such as magnetite, pyrrhotite, sphalerite, chalcopyrite, and galena (Lim et al., 2013) (Fig. 3g–i). The magnetite is replaced by sphalerite and pyrite (Fig. 3g), and chalcopyrite exsolution rarely occurs in sphalerite (Fig. 3g–i). The mineral stability relationships and chemical compositions of the skarn and sulfide minerals constrain the temperature of the main mineralization event to >300 °C (Lim et al., 2013).

4. Analytical methods

To examine the trace element concentrations of sphalerite in the various deposit types and ore species, representative ore samples (3 for Weolak, 5 for Seobo, 6 for Subok, 4 for Keumsil, 4 for Wooseok, 4 for Geumseong, 6 for Dangdu) were collected from each deposit. Ore petrographic studies were conducted on polished sections using standard reflected-light microscopy and backscattered electron (BSE) images. Special attention was paid to characterizing the relationship

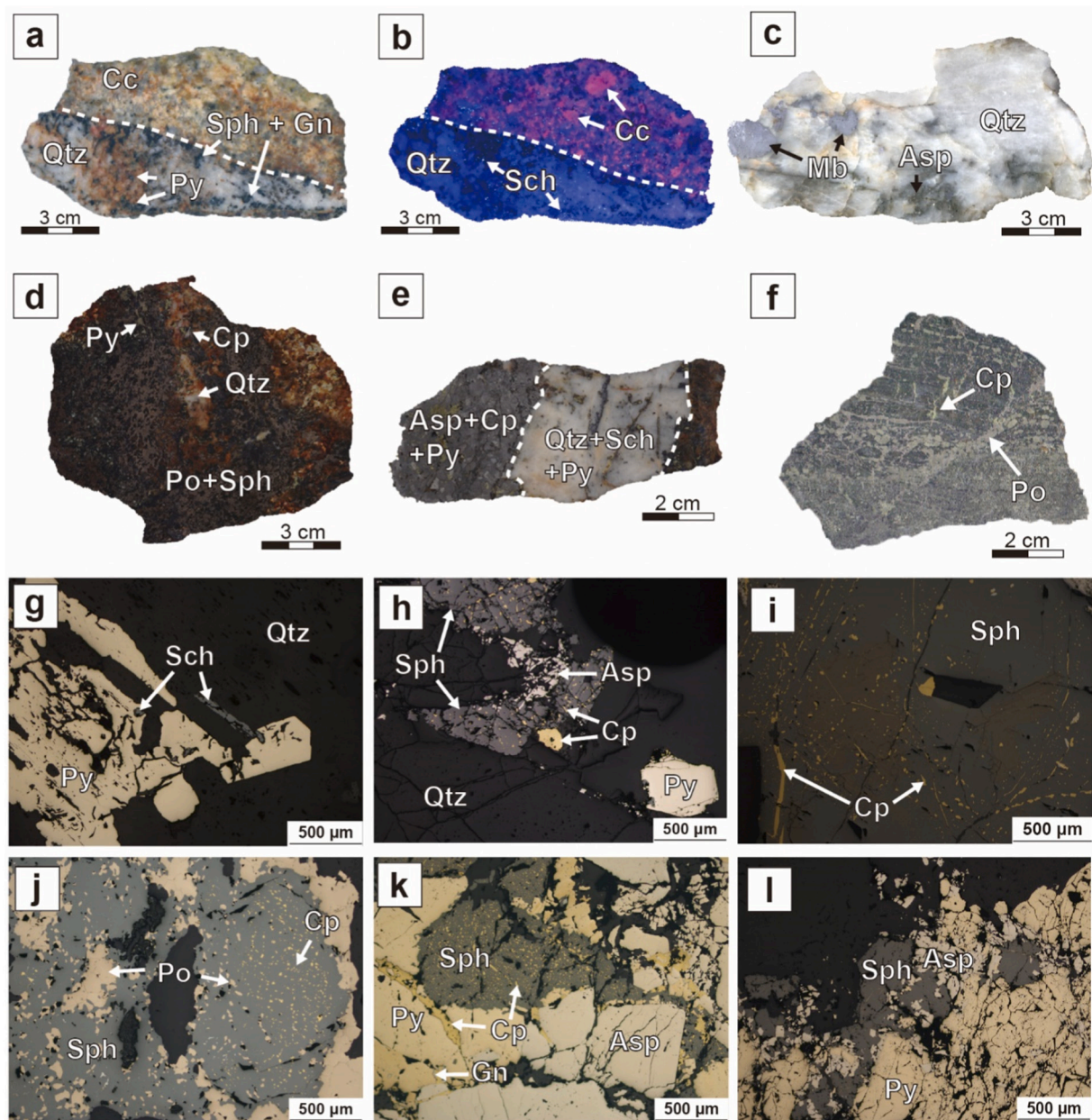


Fig. 2. Rock slaps and reflected light microscopy images of hydrothermal vein deposits in the study area. a) quartz vein accompanying sphalerite, galena, and pyrite (SEO-17), b) disseminated scheelite in quartz vein under UV lamp (SEO-17), c) quartz vein with disseminated molybdenite and arsenopyrite (WA-9), d) quartz–pyrrhotite–sphalerite assemblage with minor pyrite and chalcopyrite (WS-8), e) massive arsenopyrite–chalcopyrite–pyrite in contact with quartz vein (SB-3), f) development of pyrrhotite and chalcopyrite veinlet hosted in argillaceous sedimentary rock (KS-12), g) scheelite coexisting with pyrite in quartz vein (SEO-3), h) arsenopyrite and sphalerite with disseminated chalcopyrite in quartz vein (SEO-21), i) chalcopyrite in sphalerite forming bead chain or veinlet shape (WA-11-2), j) chalcopyrite disease in sphalerite with pyrrhotite inclusion (WS-8-2), k) Sphalerite with chalcopyrite disease replacing or infilling cracks in pyrite and arsenopyrite (SB-3-1), l) brecciated sphalerite intergrown with pyrite and arsenopyrite (KS-9). Sph: sphalerite, Py: pyrite, Asp: arsenopyrite, Cp: chalcopyrite, Gn: galena, Po: pyrrhotite, Mc: marcasite, Sch: scheelite, Mb: molybdenite, Qtz: quartz, Cc: calcite.

between chalcopyrite disease and the In–Ga concentration in sphalerite by examining the chalcopyrite exsolution texture.

To determine the major and trace element concentrations in the sphalerite, LA-ICP-MS spot analyses were conducted on 49 polished sphalerite sections using an Agilent 7700 quadrupole ICP-MS coupled with a 193 nm excimer laser ablation system (ESI NWR 193, USA) at the Korea Institute of Ocean Science and Technology, Busan. The beam diameter ranged from 15 to 50 μm, depending on the size of the sphalerite. The pulse repetition rate was 10 Hz, and the energy density was 5 J/cm². The analysis time for each point was 130 s: 50 s for the background and 80 s for the samples. The analyzed elements included

Zn, Fe, Mn, Cd, Co, Cu, Pb, Sn, In, Ga, Ge, Hg, Au, Ag, As, Sb, Ni, Te, W, Mo, Tl, V, U, Ti, and Se. External calibration for the analysis was performed using the USGS sulfide standard MASS-1 (Wilson et al., 2002), and the STDGL3 reference material (University of Tasmania) was used as an auxiliary standard for Ti, Ge, Au, and U. Zinc concentration determined by Electron Probe Microanalysis (EPMA) on each sample was used as an internal standard.

5. Results

The data of LA-ICP-MS spot analyses are summarized as the average,

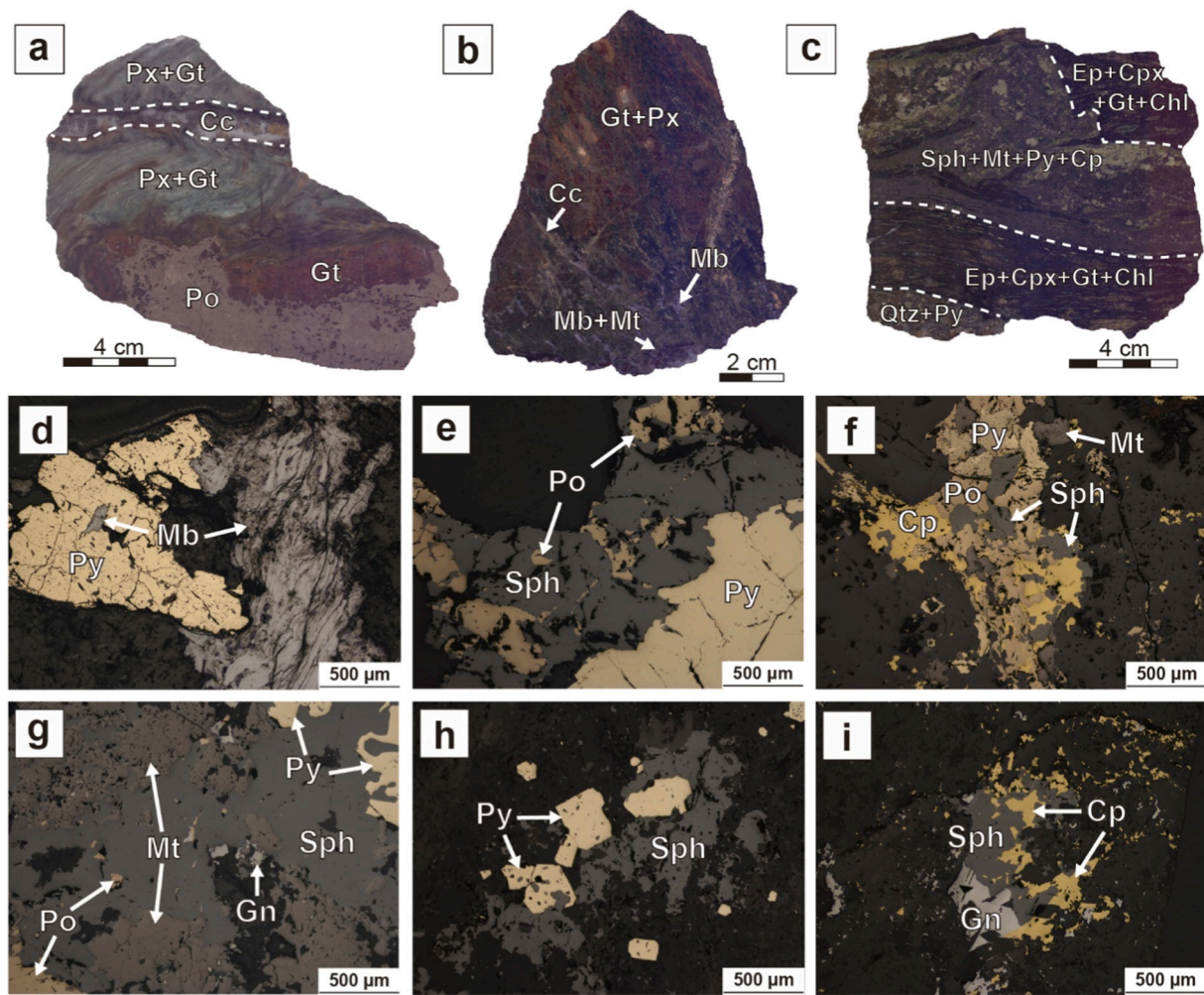


Fig. 3. Rock slaps and reflected light microscopic images of skarn deposits in the study area. a) pyroxene–garnet or garnet skarn replaced by massive pyrrhotite and calcite vein (GS-16), b) garnet–pyroxene skarn replaced by molybdenite and magnetite forming flame texture and late calcite veinlet (GS-2), c) massive ore comprising sphalerite, magnetite, pyrite, and chalcopyrite in contact with skarn assemblage consisting of epidote, clinopyroxene, garnet, and chlorite (DD-5), d) molybdenite occurring with subhedral pyrite (GS-2-2), e) pyrite and sphalerite replaced by pyrrhotite (GS-12-2), f) pyrrhotite–chalcopyrite postdating magnetite–pyrite–sphalerite (GS-17), g) magnetite and pyrite replaced by sphalerite–pyrrhotite–galena (DD-5), h) subhedral pyrite partly replaced by sphalerite (DD-6), i) sphalerite coexisting with chalcopyrite and galena (DD-10). Mt.: magnetite, Gt: garnet, Px: pyroxene, Cpx: clinopyroxene, Ep: epidote, Chl: chlorite. Refer to Fig. 2 for others.

standard deviation, minimum, and maximum for each element (Table 2), and the complete datasets are provided in the Online supplementary materials (Supplementary Table 1).

The Fe contents of sphalerite of the two W deposits (Seobo and Weolak) range from 1.8×10^4 to 6.7×10^4 ppm (average: 3.5×10^4 ppm) and from 1.1×10^4 to 10.6×10^4 ppm (average: 8.0×10^4 ppm), respectively. The former had the lowest Fe content among the studied deposits. The highest average value 9.3×10^4 ppm, ranging from 6.2×10^4 to 12.3×10^4 ppm, was recorded in the Subok W–Zn–Pb deposit. The average Mn concentrations in all the studied deposits ranged from 2146 ppm in the Geumseong deposit to 9267 ppm in the Keumsil deposit. The lowest Cd concentration was observed in the Geumseong deposit, averaging 4124 ppm, and the highest in the Weolak deposit, averaging 12.6×10^3 ppm. Copper is the most abundant trace metal detected in the sphalerite and generally varies from tens to hundreds, and even up to 1000 ppm in this study. The highest Cu concentration among the studied deposits is observed in the Woosok deposit, ranging from 107 to 1102 ppm (average: 522 ppm), and the lowest is in the Geumseong deposit, ranging from 24.8 to 104 ppm (average: 46.7 ppm).

The concentrations of the trace elements, including Co, Cu, Pb, Sn, In, Ga, Hg, Ag, As, and Sb, range from a few ppm to thousands of ppm,

whereas those of Ge, Ni, Te, W, Mo, Tl, and Ti are mostly below the detection limit (Table 2). The In concentration tends to vary widely from a few ppm to hundreds of ppm; the lowest In value is observed in the Geumseong deposit, ranging from 13.15 to 66.4 ppm (average: 30.89 ppm), and the highest in the Woosok deposit, ranging from 128 to 1825 ppm (average: 685 ppm) (Fig. 4).

Relatively small amounts of Ga are also detected in the different deposits. The highest Ga concentration (average: 14.97 ppm, range: 0.39–34.93 ppm) is found in the Geumseong deposit, while the Weolak and Dangdu deposits have very low Ga levels of 2.31 ppm and 2.28 ppm, respectively. In this study, the Ge concentrations in the sphalerite is extremely low (generally <0.1 ppm), while the Sn concentration varies considerably, its range spanning three orders of magnitude. The highest average Sn concentration (477 ppm) is measured in Keumsil, whereas both the Geumseong and Dangdu skarn deposits have average concentrations <1 ppm.

Table 2
Summary of LA-ICP-MS data for sphalerite from the deposits in the Hwanggangri mineralized district (ppm).

Deposit		Fe	Mn	Cd	Co	Cu	Pb	Sn	In	Ga	Ge	Hg	Ag	As	Sb	Ni	Te	W	Mo
Weolak (n = 12)	av.	80,031	4796	12,662	59.08	266.8	0.56	4.33	283.34	2.31	bdl	4.65	1.16	0.85	0.21	bdl	0.01	4.48	0.00
	std.	34,656	4011	2171	28.28	111.0	1.15	1.90	78.68	1.03	bdl	3.95	1.09	2.94	0.57	bdl	0.03	15.51	0.01
	min	10,817	80	10,215	bdl	146.8	bdl	1.18	179.05	0.86	bdl	1.71	0.36	bdl	bdl	bdl	bdl	bdl	bdl
	max	106,418	9541	16,976	86.31	536.7	3.99	8.79	401.44	3.73	bdl	12.08	4.20	10.20	2.00	bdl	0.09	53.74	0.02
Seobo (n = 20)	av.	34,505	2681	8883	1.74	63.4	0.78	4.14	67.37	3.52	0.01	8.75	1.38	4.99	0.45	0.01	0.06	0.00	0.00
	std.	11,971	2467	2760	2.68	66.5	0.83	6.97	84.14	2.43	0.03	3.03	0.78	19.44	0.68	0.05	0.15	0.00	0.02
	min	18,338	603	4700	bdl	4.7	bdl	0.42	1.97	1.25	bdl	4.05	0.38	bdl	bdl	bdl	bdl	bdl	bdl
Subok (n = 34)	max	66,915	8432	14,920	7.41	237.9	3.35	26.37	300.27	9.44	0.15	13.98	3.26	87.50	2.65	0.22	0.56	0.01	0.07
	av.	93,392	3953	10,629	20.25	274.7	5.03	23.46	322.37	6.60	0.00	2.07	2.71	1.39	0.07	bdl	0.02	0.01	0.00
	std.	12,947	2314	1938	20.84	158.8	27.21	32.84	199.62	3.97	0.01	0.87	2.26	7.68	0.26	bdl	0.10	0.03	0.01
Keumgil (n = 22)	min	62,071	70	7578	0.03	95.2	0.00	1.58	74.35	1.13	bdl	0.82	0.85	bdl	bdl	bdl	bdl	bdl	bdl
	max	123,766	8372	14,518	68.45	760.1	158.99	157.90	855.27	16.56	0.05	5.13	13.25	44.86	1.18	bdl	0.52	0.20	0.04
	av.	46,999	9267	11,794	2.66	145.9	36.91	66.76	103.99	3.94	0.01	13.52	6.85	4.13	3.52	bdl	0.09	0.00	0.23
	std.	18,353	10,006	7503	4.96	143.0	90.29	150.25	80.35	3.14	0.03	9.43	11.14	12.03	5.95	bdl	0.30	0.00	0.54
Wooseok (n = 15)	min	19,964	579	4107	bdl	29.8	0.00	1.84	4.98	bdl	bdl	3.42	0.12	bdl	bdl	bdl	bdl	bdl	bdl
	max	68,991	25,331	26,524	20.33	630.5	335.87	557.63	287.57	10.27	0.12	32.68	52.47	54.61	26.49	bdl	1.41	0.02	2.29
	av.	74,963	3479	11,842	1.91	522.3	138.97	5.98	695.87	3.89	bdl	4.16	7.92	0.27	0.67	bdl	0.04	0.49	0.00
	std.	8408	2959	2128	1.49	308.9	523.00	9.16	450.47	4.07	bdl	2.16	6.27	0.64	1.42	bdl	0.09	1.87	0.01
Geumseong (n = 49)	min	55,437	434	7075	0.03	107.2	0.11	0.93	128.29	0.45	bdl	1.74	1.31	bdl	bdl	bdl	bdl	bdl	bdl
	max	90,250	8270	14,318	4.95	1101.9	2029.31	37.02	1825.02	12.70	bdl	8.82	25.74	2.26	5.10	bdl	0.30	7.26	0.04
	av.	71,444	2146	4124	1.35	46.7	3.20	0.80	31.60	15.56	0.01	3.84	2.61	0.05	0.71	0.01	0.06	0.00	0.00
	std.	4121	1102	460	2.47	16.0	2.41	0.95	18.46	9.61	0.03	0.79	1.86	0.16	0.91	0.09	0.10	0.00	0.00
Dangdu (n = 52)	min	61,054	531	3451	0.00	24.8	0.53	0.23	15.05	1.93	0.00	2.21	0.71	bdl	0.04	bdl	bdl	bdl	bdl
	max	80,418	4038	5115	8.65	104.2	13.13	6.41	66.40	34.93	0.16	5.12	9.35	0.79	4.72	0.64	0.62	0.01	0.01
	av.	80,779	7759	11,851	52.97	59.2	2.33	0.30	42.42	2.28	0.01	1.70	1.07	0.06	0.04	0.12	0.01	0.00	0.02
	std.	15,240	4138	2301	78.15	55.7	5.48	0.22	34.41	2.83	0.03	0.78	0.85	0.21	0.11	0.26	0.04	0.01	0.09
	min	42,035	1662	7985	0.05	12.3	bdl	0.12	5.20	0.12	bdl	0.72	0.16	bdl	bdl	bdl	bdl	bdl	bdl
	max	98,378	15,748	16,010	340.09	320.9	30.89	1.34	134.14	11.90	0.12	4.39	3.85	1.21	0.61	0.92	0.22	0.04	0.65

bdl = below the detection limit.

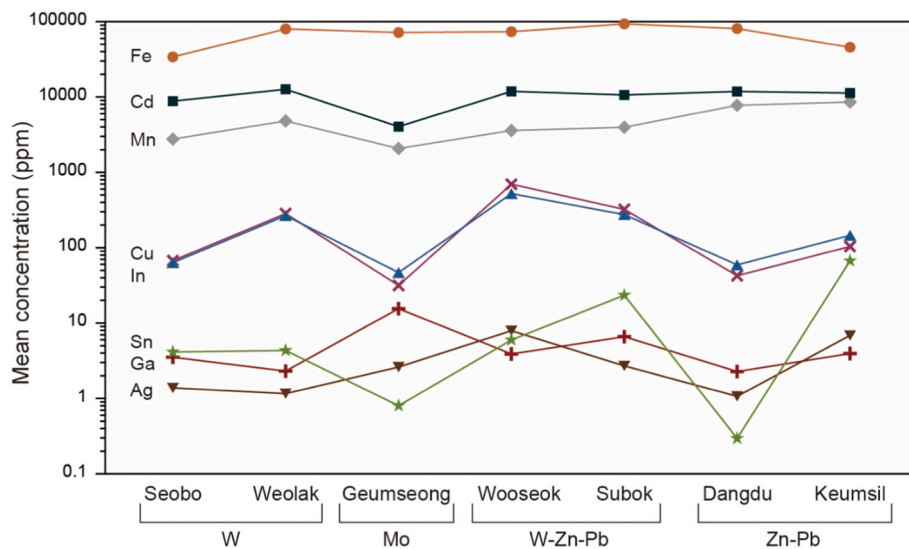


Fig. 4. Comparison of average concentrations of selected elements in sphalerite determined by LA-ICP-MS for each deposit.

6. Discussion

6.1. Elemental behavior in sphalerite

The concentrations of trace elements, such as In, Ga, Cu, and Hg, as well as major and minor elements, including Zn, Fe, and Mn, were uniformly measured in sphalerite free of inclusion minerals (Fig. 5). However, sphalerite with an inhomogeneous texture can produce irregular signals, which can be attributed to the presence of subsurface inclusions (Fig. 6). Typically, in sphalerites with chalcopyrite disease from the Wooseok and Subok deposits, Cu shows irregular signals (Fig. 6c, f), and the concentrations of Sn and Ag vary similarly to that of Cu (Fig. 6c). This suggests the preferential incorporation of Sn and Ag into chalcopyrite over sphalerite when these minerals are formed. Significant irregular signals are observed in the sphalerite occurring with pyrrhotite, pyrite, and galena from the Dangdu deposit. For instance,

abnormal peaks of Ti, Ag, and Sn are detected in sphalerite that is rich in pyrrhotite inclusions (Fig. 6g–i). However, relatively homogeneous concentrations of In and Ga are measured (Fig. 6c, f), indicating that these two elements are present primarily in the sphalerite and are not present in the other inclusion minerals described above.

In sphalerite, trace elements are substituted for Zn and Fe that exist in the lattice structure, and although substitution sometimes occurs between single elements, they are commonly combined with other elements for substitution (Johan, 1988; Cook et al., 2009; Ye et al., 2011; Murakami and Ishihara, 2013; Belissont et al., 2014; Wei et al., 2019). Elements with divalent cations, such as Fe, Mn, Cd, and Co, mainly substitute Zn, whereas elements with monovalent, trivalent, or tetravalent cations, such as Ag^+ , In^{3+} , and Sn^{4+} , are combined to replace Zn and Fe in accordance with their chemical properties (i.e., ionic bonds and radii) (Cook et al., 2009; Murakami and Ishihara, 2013).

Thus, the interrelationships and distribution of the trace elements in

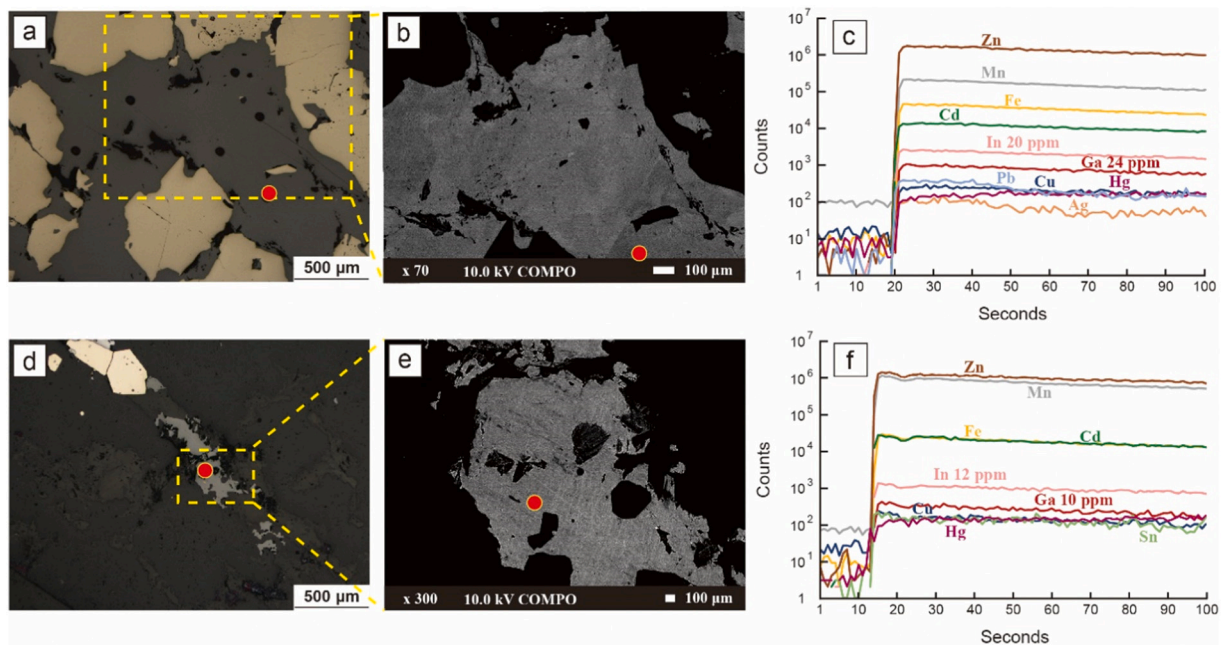


Fig. 5. Reflected light microscopic and backscattered electron (BSE) images of sphalerites without inclusions and associated irregular time-resolved signals obtained by LA-ICP-MS. a–c) Geumseong, d–f) Keumstil deposit.

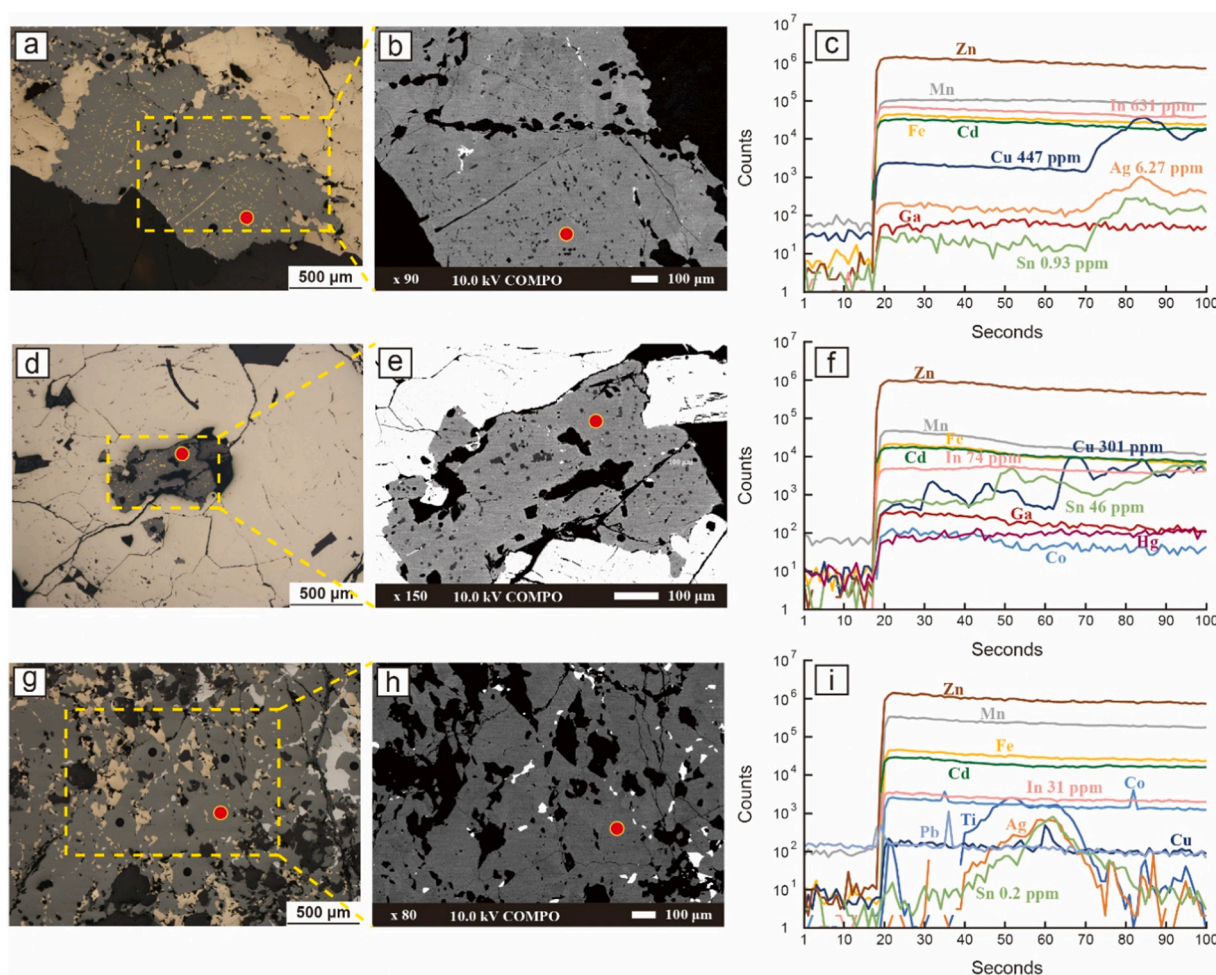


Fig. 6. Reflected light microscopic and backscattered electron (BSE) images of sphalerites containing inclusions and associated irregular time-resolved signals obtained by LA-ICP-MS. a–c) Wooseok, d–f) Subok, g–i) Dangdu deposit.

the sphalerite were analyzed and compared. Indium and Cu are preferentially concentrated on the (111) plane in sphalerite and participate in 1:1 substitution via bonding, as described in the substitution formula $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{In}^{3+}$ (Johan, 1988). In the study area, the majority of the In and Cu displays a linear tendency with a 1:1 ratio, suggesting that they both participate in substitution (Fig. 7a). However, in some cases, submicron inclusions of chalcopyrite cause elevated Cu and depleted In below the In/Cu line of ~1 (Cook et al., 2009; Ye et al., 2011; Murakami and Ishihara, 2013). Furthermore, Sn exhibits a weak positive correlation with Cu (Fig. 7b). However, Cu data that are far below the 1:1 line suggest that Cu substitutes Zn more readily than Sn. The Ag content of sphalerite also displays a positive correlation with Cu, although the correlation differs among the deposits (Fig. 7c). The Ag:Cu ratios are mostly <1, indicating that Cu is more favorably substituted than Ag. The Ag data are close to the 1:1 line of Ag:Sn, suggesting that these two elements exhibit similar behavior during substitution (Fig. 7d).

These relationships suggest that Cu shows a positive correlation not only with In but also with other elements such as Sn and Ag. This raises the possibility of coupled substitutions, $3\text{Zn}^{2+} \leftrightarrow 2(\text{Cu}, \text{Ag})^+ + \text{Sn}^{4+}$ in addition to $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{In}^{3+}$. These coupled substitutions would account for the behavior of Cu, Ag, and Sn in the sphalerite from the study area (Cook et al., 2009; Ye et al., 2011; Murakami and Ishihara, 2013) across a wide range of In concentrations. This interpretation is supported by the fact that In tends to behave incompatibly with other elements during granite fractionation (Simons et al., 2017), and exhibits behavior similar to that of Cu–Sn–Ag as a chalcophile element, establishing a high correlation. However, it should be noted that the elements

in the sphalerite from the Geumseong and Dangdu skarn deposits deviate significantly from the positive correlation lines for the paired elements, particularly those for Sn–Cu, Ag–Cu, and Ag–Sn (Fig. 7b–d). It appears that the compositions of the sphalerite in these two deposits are more strongly controlled by a single coupled substitution, $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{In}^{3+}$. Because the content and behavioral characteristics of elements in sphalerite differ depending on the genetic environment and the involved fluids during deposition, deposits with values that do not form linear relationships may form under different conditions. However, both the Geumseong and Dangdu deposits are representative skarn deposits, and it is presumed that local variations in the composition of the host rock and infiltrating magmatic fluids controlled the elemental behavior in these deposits.

Previous studies have suggested that Ag and Sb, in the form of Ag^+ and Sb^{3+} ions, participate in combined substitution (Cook et al., 2009; Lockington et al., 2014). Considering that the Ag and Sb data in this study tend to be concentrated subparallel to the Ag/Sb = 1 correlation line (Fig. 7e), it is inferred that substitution occurred according to $2\text{Zn}^{2+} \leftrightarrow \text{Ag}^+ + \text{Sb}^{3+}$. However, the Ag:Sb ratios are mostly >1, indicating that Ag is more active in the substitution than Sb, or that micro- or nano-scale inclusions of Ag-rich sulfosalts may exist within the sphalerite (Cook et al., 2009).

Gallium and In both exhibit uniform downhole profiles in the sphalerite, suggesting that they are commonly hosted in solid solution within this mineral (Fig. 6c, f). Both elements may exist in a trivalent form and participate in coupled substitution with monovalent cations, particularly Cu^+ (Schwarz-Schampera and Herzog, 2002; Shimizu and

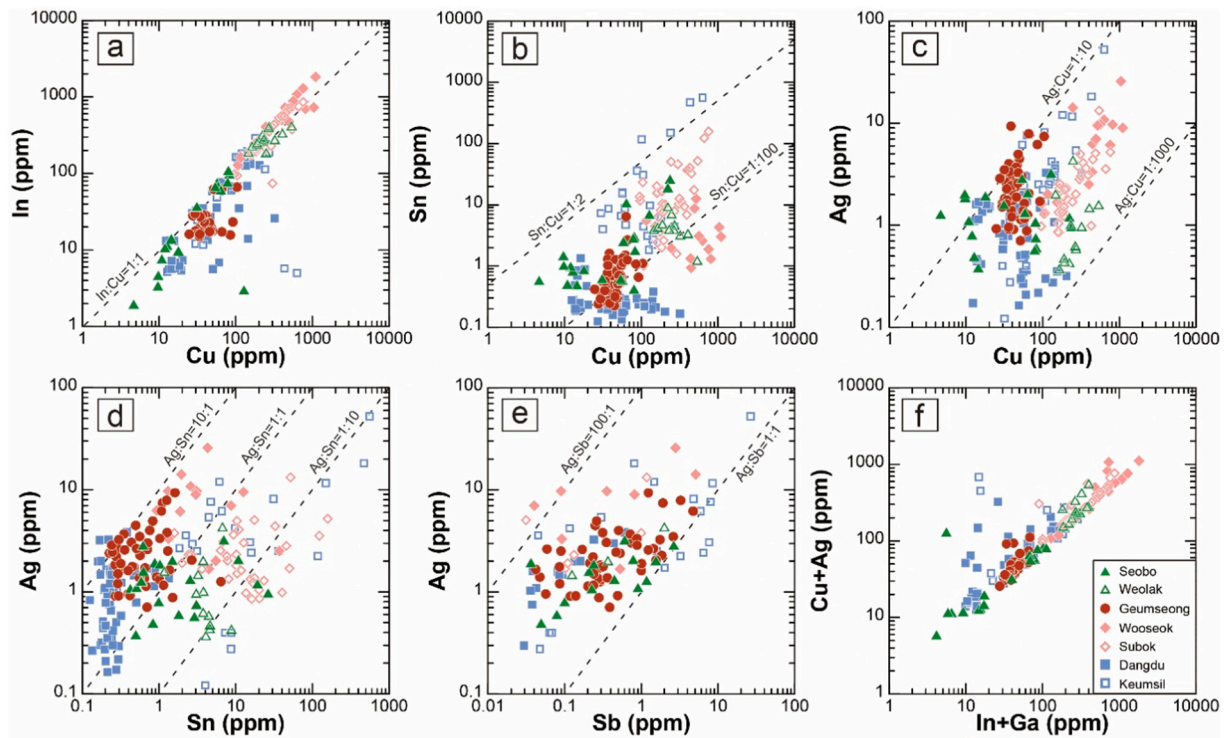


Fig. 7. Binary plots for selected trace elements in sphalerite obtained by LA-ICP-MS showing the substitution relationship between the elements. a) Cu vs. In, b) Cu vs. Sn, c) Cu vs. Ag, d) Sn vs. Ag, e) Sb vs. Ag, f) In + Ga vs. Cu + Ag.

Morishita, 2012; Bonnet et al., 2016; Sahlström et al., 2017). The Cu + Ag vs. In + Ga plot exhibits a broadly positive correlation (Fig. 7f), supporting the occurrence of the coupled substitution $2\text{Zn}^{2+} \leftrightarrow (\text{Cu},$

$\text{Ag})^{+} + (\text{In}, \text{Ga})^{3+}$. However, the Cu + Ag: In + Ga ratio, which is close to or exceeds 1, may be because the Cu and Ag substitution was more favorable than that of In and Ga.

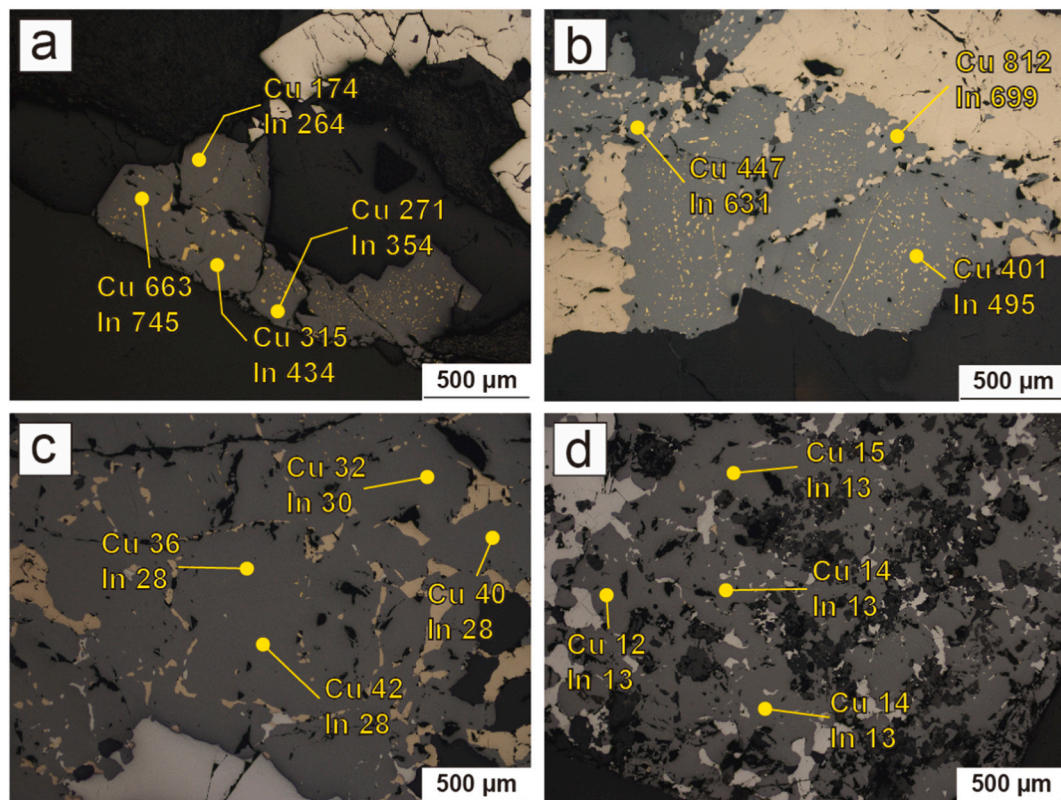


Fig. 8. Reflected light microscopic images showing the concentrations of Cu and In in sphalerite with and without chalcopyrite disease. a–b) Subok and Woosok deposits with chalcopyrite disease, c–d) Geumseong and Dangdu deposits without chalcopyrite disease. Unit in ppm.

6.2. Indium associated with chalcopyrite disease

Chalcopyrite inclusions commonly observed in sphalerite are referred to as “chalcopyrite-disease”. Several mechanisms have been proposed to explain the origin of the texture; exsolution through unmixing from a high temperature sphalerite (Ramdohr, 1980), replacement of Fe by Cu-bearing fluids (Barton and Bethke, 1987; Eldridge et al., 1988; Bortnikov et al., 1991), and co-precipitation (Kojima, 1990). However, the first mechanism has been discarded on the basis of experimental studies in the Cu-Fe-Zn-S system (Wiggins and Craig, 1980; Hutchison and Scott, 1981; Kojima and Sugaki, 1985). Chalcopyrite can form a linear array within the sphalerite or may be distributed as irregular spots (Fig. 8). In this study, the development and frequency of chalcopyrite disease differ between deposits. Most sphalerites from the W deposits of Seobo and Weolak and the W–Zn–Pb deposits of Wooseok and Subok show abundant chalcopyrite disease. In the Wooseok and Subok deposits, the sphalerite shows a higher In and Cu contents than in other deposits, with a distinct proportional relationship (Fig. 7a). There occurs a significant variation in these concentrations even within a single crystal; for example, 174–663 ppm Cu and 264–745 ppm In in Subok samples and 401–812 ppm Cu and 495–699 ppm In in Wooseok samples (Fig. 8a, b).

In contrast, chalcopyrite commonly forms composite grains with sphalerite, pyrrhotite, or galena, displaying a random distribution in various sizes in the Geumseong Mo and Dangdu Zn–Pb deposits (Fig. 3f, i). The poor development of chalcopyrite disease in these deposits can be attributed to either a restricted supply of additional Cu-bearing fluids or the coprecipitation of sphalerite and chalcopyrite, featuring lower concentrations of Cu and In, such as 32–42 ppm Cu and 28–30 ppm In for the Geumseong deposit, and 12–16 ppm Cu and 13 ppm In for the Dangdu deposit (Fig. 8c, d).

According to Cook et al. (2012), In and Cu initially formed solid solutions with In substituting for Cu in Cu–In-bearing sphalerite, establishing a strong correlation between these two elements. This correlation is similarly illustrated in the results of the present study (Fig. 7a). Subsequently, the Cu became separated owing to environmental changes, such as cooling, resulting in chalcopyrite disease, while In remained in the sphalerite (Kieft and Damman, 1990). It has been previously reported that Fe-rich sphalerites with chalcopyrite disease exhibit high In concentrations (Bortnikov et al., 1991; Schwarz-Schampera and Herzig, 2002; Seifert and Sandmann, 2006). Therefore, the presence and development of chalcopyrite disease can potentially be used to estimate In enrichment in sphalerite, with strong implications in the context of mineral exploration (Johan, 1988; Cook et al., 2012; Frenzel et al., 2016a).

6.3. Variations in sphalerite composition based on ore species and deposit type

Sphalerite that crystallizes from different deposits exhibits distinct chemical compositions that reflect the genetic environment in which they form. The major and trace element compositions of sphalerite have been analyzed to classify ore deposits according to deposit type or ore species, interpret deposit-forming environments, and serve as geochemical proxies for exploration (Chon and Shimazaki, 1986; Choi, 1993; Kelley and Leach, 2004; Ye et al., 2011; Lockington et al., 2014; Im et al., 2018; Yuan et al., 2018a; Bauer et al., 2019; Im et al., 2022).

In sphalerite from various metallic ore deposits, it is common to find Mn (up to 2 wt% or more) (Graeser, 1969; Sombuthawee et al., 1978; Gottesmann et al., 2009; Lee et al., 2019; Damian et al., 2021; Dincă et al., 2022). In this study, the Mn concentrations are comparatively high in the Zn–Pb deposits: 7759 ppm for the Dangdu deposit and 9267 ppm for the Keumsil deposit, whereas the Geumseong Mo deposit has the lowest value (2146 ppm) (Fig. 4). The sphalerite in the Zn–Pb deposits in the Taebaeksan region is also enriched with Mn, up to 2.63 wt %, but is depleted in Cd (Chon and Shimazaki, 1986; Lee et al., 2019).

Sphalerite from deposits associated with igneous activity, such as skarn and volcano-hosted massive sulfide deposits, is relatively enriched in Mn (several thousands of ppm or above), whereas that from Mississippi Valley type (MVT) deposits is depleted in Mn (several tens of ppm or below) (Cook et al., 2009; Ye et al., 2011; Frenzel et al., 2016a). Therefore, the high Mn concentrations (up to 2.5 wt%) in the Dangdu and Keumsil deposits are attributed to magmatic-hydrothermal processes during their Zn–Pb mineralization. Based on statistical studies of Mn and Cd concentrations in sphalerite from various Au–Ag deposits in South Korea, Choi (1993) demonstrated that high Mn concentrations are clearly distinct in epithermal Au–Ag deposits, but not in hypothermal Au deposits. The difference was attributed to the more evolved mineralizing fluids and relatively lower formation temperatures of the former deposits. The low Mn concentrations of the Geumseong Mo and Weolak W deposits, despite their magmatic-hydrothermal origin, thus could be ascribed to the relatively high formation temperatures of the ore minerals (Table 1).

The average Cd content of the sphalerite from the Zn–Pb or W–Zn–Pb deposits ranges from 1.06×10^4 to 1.19×10^4 ppm, and is higher than that of the Mo skarn deposit, which averaged 4124 ppm (Fig. 4b, c). Skarn deposits are generally formed by differentiated magmatic fluids, implying that their low Cd concentrations originate from low concentrations of mineralizing fluids derived from magma (Chon and Shimazaki, 1986; Choi et al., 2007). However, Cd concentrations in the sphalerite of Zn–Pb deposits can vary depending on the deposit type (carbonate-hosted veins: 7300 ppm; low-carbonate veins: 4100 ppm; skarn deposits: 3500 ppm) (Schwartz, 2000). Lee et al. (2019) also reported that the sphalerite of carbonate replacement ores from the Taebaeksan region contains high Cd concentrations of up to 1.2 wt% (average: 7200 ppm), suggesting that this enrichment is caused by the leaching of Cd from the carbonate host rock by hydrothermal fluids. Nonetheless, the Cd content in limestone is generally low (~ 0.03 ppm) (Gong et al., 1977), although exceptionally high concentrations, up to 2.4 ppm, have been reported from Paleozoic limestone in Japan (Aizawa and Akaiwa, 1992). Therefore, it appears that the Cd content in sphalerite is controlled by various factors, such as the host rock type, magmatic-hydrothermal fluids, and physicochemical conditions, including temperature, pH, total reduced S, and ligand activity (Wen et al., 2016).

In the present study, distinct correlations between In and Ga in relation to Cd are not evident. However, a discernable weak positive correlation is observed between In and Cd, particularly in samples from W-bearing deposits, such as the Seobo deposit (Fig. 9a, b). This finding implies that the Cd content in sphalerite may potentially serve as an exploration proxy for In mineralization in some cases. Specific correlations between Fe, Mn, Cu, In, and Ga are not evident in the sphalerite (Fig. 9c–h), except for the relationship between Cu and In (Fig. 7a).

6.4. Mineralization temperature and In–Ga concentration

In the sphalerite from the Zn–Pb and W–Zn–Pb deposits (Keumsil, Dangdu, Subok, and Wooseok), where Zn is the main commodity, the In concentrations range from 42 to 685 ppm. Given that an average In content in the ores from global In-producing mines is typically <100 ppm (Roskill, 2010; Lokanc et al., 2015), the ores from these deposits are considered to hold economic potential as In deposits, provided sufficient Zn ores are secured.

The W–Zn–Pb (Subok, Wooseok) and W–Mo (Weolak) deposits of hydrothermal vein type show high In concentrations, averaging up to 696 ppm (Table 2). Globally, significant In-bearing mineralization is found in high-temperature polymetallic vein- and greisen-type deposits associated with Sn–W–Cu (e.g., Schwarz-Schampera and Herzig, 2002; Sinclair et al., 2005; Ishihara et al., 2011). Notably, major In deposits in Japan, like Toyoha (average: 300 ppm) and Ashio mines, are sub-volcanic, forming Sn-polymetallic veins (Ishihara et al., 2006). The Freiberg district in Germany, renowned for In mineralization,

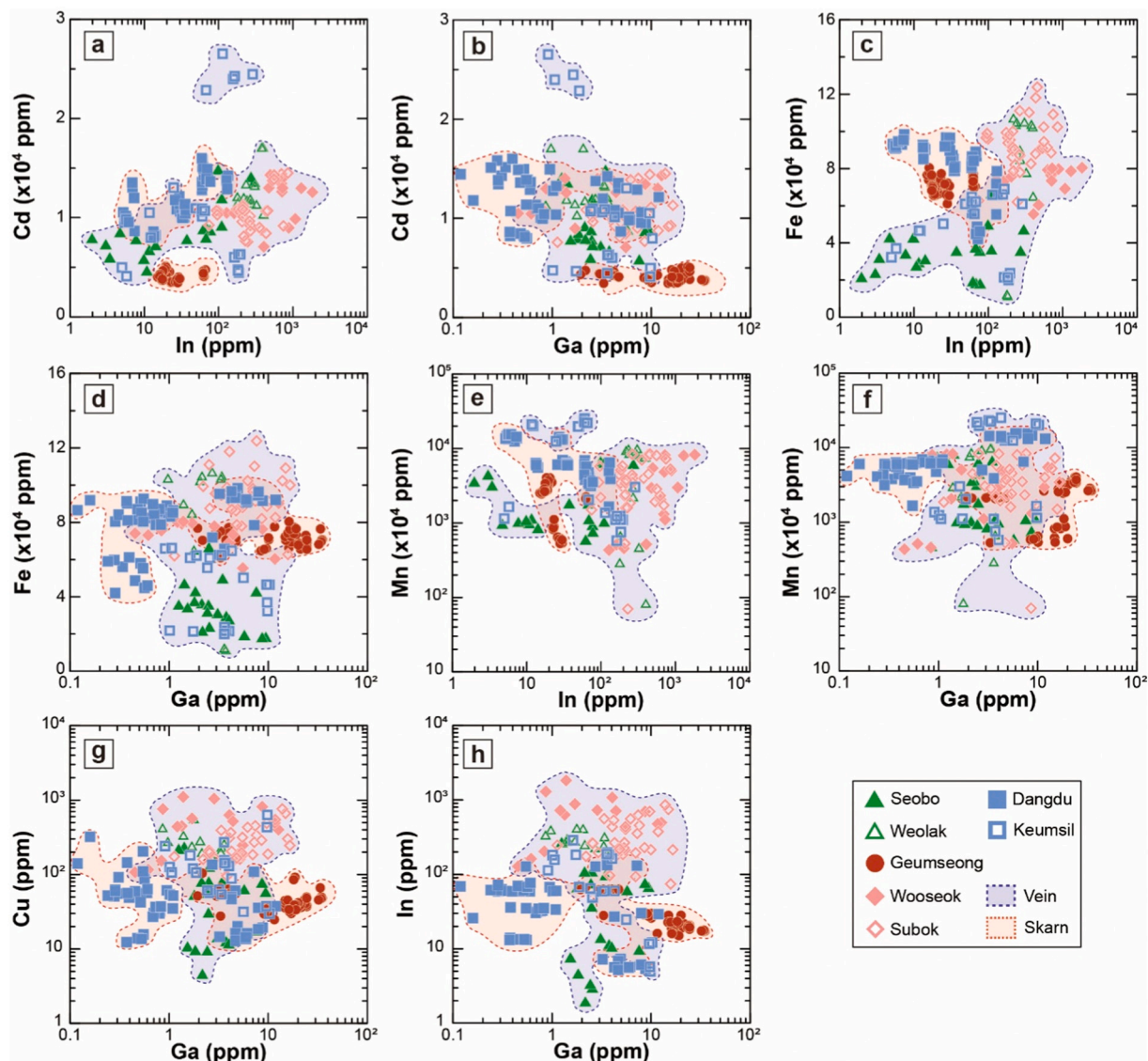


Fig. 9. Correlations of In, Ga, Cd, Fe, Mn, and Cu concentrations in the sphalerite of each deposit and their composition ranges filled with different colors depending on the deposit type of skarn and hydrothermal vein.

encompasses numerous polymetallic base metal vein deposits and granite-related Sn–W–base metal deposits (Seifert and Sandmann, 2006). The correlation between In and Sn is evident in various deposits, such as the Neves-Corvo deposit in the Iberian Pyrite Belt (Carvalho et al., 2018). The study area's sphalerite further supports positive correlations among In, Cu, and Sn (Figs. 4 and 7a–b).

Additionally, W mineralization, associated with differentiated igneous rocks, generally occurs at temperatures >300 °C (Meinert, 1995; Breiter et al., 1999; Yuan et al., 2018b). Under such high-temperature conditions, sphalerite tends to incorporate a high content of In (Cook et al., 2009; Ye et al., 2011; Murakami and Ishihara, 2013; Xu et al., 2021). Fluid inclusions in sphalerite, along with experimental results for In-bearing hydrothermal solutions, provide that the crystallization and enrichment of In occur at temperatures of 300–400 °C or higher (Seward et al., 2000; Shimizu and Morishita, 2012). Previous studies confirmed high-temperature mineralization (up to 380 °C) of In-enriched W and W–Zn–Pb deposits in the study area (Table 1).

The In content in sphalerite of the Geumseong Mo and Dangdu Zn–Pb skarn deposits is <50 ppm (Table 2), contrasting with previous skarn studies reporting higher In enrichment (Cook et al., 2009; Ye et al., 2011). In the study area, the high-temperature W and W–Zn–Pb vein

deposits have elevated In concentrations. Lee et al. (2019) also reported a high In content (322 ppm) in the Sangdong W vein deposit hosted by Paleozoic sedimentary rocks in the Taebaeksan mineralized district (TMD). However, low In concentrations (<40 ppm) were reported in various skarn (e.g., the Uljin Zn–Pb and Sagok Zn–Pb deposits) and carbonate replacement (e.g., the Wondong Fe–Zn–Pb deposit) deposits, even at a high temperature (350 to 460 °C) (Hwang and Lee, 1998). Additionally, it is known that skarn and MVT deposits in South China contain much lower In concentrations (<0.1 ppm) compared to sedimentary exhalative and volcanogenic massive sulfide deposits (>100 ppm) (Ye et al., 2011).

It is recognized that the fractionation of specific elements within sphalerite is governed by the source of metal, as well as factors such as crystallization temperature and fluid salinity, leading to variable concentrations of trace elements across deposit types or individual deposits (Cook et al., 2009; Frenzel et al., 2016a; Lee et al., 2022). The concentrations of In in granite and carbonate rock are typically low, ranging from ten to 500 ppb (Schwarz-Schampera and Herzig, 2002, and references therein). It is also reported that In concentrations in Cretaceous intrusions and Paleozoic carbonate rocks in the TMD are generally below detection limits (<0.2 ppm; Lee et al., 2022). Nonetheless, the

existence and enrichment of In in sphalerite from the HMD suggest a significant concentration of In through magmatic-hydrothermal ore-forming process under high-temperature conditions (Audétat and Zhang, 2019). As In-rich magmatic fluids interact more with host rocks, the remaining In gradually decreases. Therefore, In-rich vein deposits may form when high-temperature magmatic fluids prevail with limited interaction with host rocks. Conversely, it could be depleted in skarn and replacement-type deposits, where the magmatic fluids are substantially affected by interaction with host rocks.

The Geumseong Mo deposit exhibits the highest Ga concentration at 15.56 ppm in the HMD, surpassing an average of 6.6 ppm in other deposits (Table 2). In the TMD, the Janggun Zn–Pb deposit has the highest Ga content at 62.38 ppm, while most W and W–Zn–Pb deposits show an average Ga content of <3 ppm (Lee et al., 2019). Gallium shows a higher content than In in continental crust rocks, with averages of 16–35 ppm, 4 ppm, and 19 ppm in granite, carbonate rock, and shale, respectively (Burton and Culkin, 1978; Mielke, 1979). According to Taylor and McLennan (1995) and Rudnick and Gao (2003), the average Ga concentration of the upper continental crust ranges from 17 to 17.5 ppm. Similar Ga concentrations are observed from granitic rocks (Muamsa and Weolaksan) at 17.7–37.3 ppm in the HMD (Lee et al., 2010), and from granitoid intrusions (13–17 ppm) and sedimentary sequences (up to 24 ppm) comprising limestone, sandstone, and shale in the TMD (Lee et al., 2022). Therefore, the Ga concentrations in sphalerite from the study area, lower than those in the granitic and carbonate rocks, result from factors other than the host rock composition. In sphalerite, In and Ga concentrations correlate with mineralization temperature. Specifically, In is strongly linked to high-temperature magmatic-hydrothermal systems, while Ga is generally favored in low temperature conditions (Cook et al., 2009; Shimizu and Morishita, 2012; Murakami and Ishihara, 2013; Frenzel et al., 2016a, 2016b). For example, carbonate-hosted low temperature Zn–Pb sulfide deposits, such as MVT deposits, exhibit Ga concentrations ranging from several tens to hundreds, even up to thousands of ppm in sphalerite, with homogenization temperatures between 75 °C and 200 °C (Cook et al., 2009; Leach et al., 2010; Foley et al., 2017). Lee et al. (2022) noted spatio-temporal variation in In–Ga concentrations in sphalerite from a lens-shaped Zn–Pb carbonate replacement orebody in the TMD with the highest In concentration at the center and decreasing concentrations toward the periphery, while Ga concentrations showed the opposite trend.

6.5. Fluid characteristics related to In–Ga enrichment

The sulfur isotope compositions of In-enriched Sn-polymetallic deposits (e.g., Akenobe and Fukoku in Japan and Erzgebirge in Germany) (−3 to +3 ‰) suggest magmatic origin for In mineralization (Jung and Seifert, 1996; Seifert, 1999; Schwarz-Schampera and Herzig, 2002). Murakami and Ishihara (2013) proposed a relation between In enrichment and sulfur-rich oxidized magma based on their study of In-bearing sphalerite from polymetallic deposits in Bolivia, China, and Japan. It is also notable that magmatic volatiles, including fluorine and chlorine, contribute to high concentrations of metal ligands (e.g., InCl_4^- and InClOH^+), supporting the magmatic influence on In, Sn, Se, and Bi ore minerals (Seward et al., 2000).

The Cretaceous intrusions of the Muamsa, Susan, and Weolak granites in the HMD are highly evolved potassic granites of magnetite series and I-type characteristics (So and Yun, 1992, 1994). These granites are enriched by a factor of at least two in Li, Rb, F, W, and Pb compared to average granite and other Late Cretaceous granites in the southern Korean Peninsula (Tsusue et al., 1981; Chon, 1983; Jin, 1988). Sulfur isotope studies on sulfides produced $\delta^{34}\text{S}$ values of 0.9 to 7.6 ‰ in the W-bearing deposits (Weolak, Seobo, Subok, and Wooseok) and of 5.6 to 9.9 ‰ in the Keumsil Pb–Zn deposit (So et al., 1983; So and Yun, 1992, 1994; Im et al., 2018, 2022). It indicates a magmatic source of sulfur for the W-bearing deposits, while a partially mixed sulfur with an isotopically heavier source, possibly sulfates in the sedimentary rocks, for the

Zn–Pb mineralization. These isotope characteristics align with the behavior observed during In mineralization associated with the magma intrusions discussed above.

It is explained that the temperature dependence of In and Ga concentrations in hydrothermal fluids is related to their prevalent ion complexes. Indium forms chloride complexes (e.g., InCl_3) at temperatures up to 350 °C (Seward et al., 2000), while Ga remains stable as hydroxide complexes (e.g., $\text{Ga}(\text{OH})_3$) at temperatures <250 °C (Wood and Samson, 2006). Low In and high Ga concentrations were observed in the Angouran Zn–Pb deposit, Iran, where sphalerite precipitated from a low temperature (< 200 °C) hydrothermal system with $\delta^{34}\text{S}$ values, 5.1 to 10.9 ‰, which is atypical of magma-derived sulfur (Zhuang et al., 2019). Nonetheless, any notable correlations between In and Ga concentrations are not observed in the samples of the present study (Fig. 9h).

Despite the skarn Mo deposit having the highest Ga concentration of 34.9 ppm, associated with relatively high temperatures, sphalerites in the HMD deposits are characterized by comparatively low Ga but elevated In concentrations. These are likely attributed to the high temperatures of the mineralizing fluids, which consistently exceed 300 °C across all types of deposits (Table 1), favoring chloride complexes (such as InCl_3) over hydroxide complexes (such as $\text{Ga}(\text{OH})_3$). However, local contribution of host rock composition may also influence the Ga concentration in the skarn Mo deposit.

7. Conclusions

LA-ICP-MS analysis of sphalerite samples from seven metallic ore deposits in the HMD reveals that the hydrothermal W and W–Zn–Pb vein deposits are enriched in In up to 696 ppm on average. In contrast, the Mo and Zn–Pb skarn deposits have low In concentrations, averaging <43 ppm. Similarly, the Ga content is relatively low, with an average value below 16 ppm.

Sphalerite with abundant chalcopyrite disease exhibits elevated concentrations of Cu and In, revealing a strong positive correlation between these two elements. The binary correlations between the trace elements suggest that fluids enriched in Cu, Sn, and Ag also transported In, raising the possibility of coupled substitution in sphalerite, such as $3\text{Zn}^{2+} \leftrightarrow 2(\text{Cu}, \text{Ag})^+ + \text{Sn}^{4+}$ and $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{In}^{3+}$. Additionally, In shows a weak positive correlation with Cd, mostly in the W-bearing deposits, indicating that Cd could be a useful exploration proxy in some cases.

The high In and low Ga concentrations in the sphalerite from the HMD suggest that the mineralizing fluids are relatively high in temperature (> 250 °C). This is related to the prevalence of In-chloride complexes in high-temperature fluids and Ga-hydroxide complexes in low-temperature fluids. The enrichment of In in hydrothermal vein ores compared to skarn ores emphasizes the direct contribution of magmatic fluids to ore mineralization under high temperature conditions with little influence from the host rock.

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CRedit authorship contribution statement

Hyewon Kim: Investigation, Methodology, Writing – original draft. **Dongbok Shin:** Conceptualization, Funding acquisition, Supervision, Writing – review & editing. **Heonkyung Im:** Investigation, Methodology. **Byeongyong Yu:** Software, Visualization. **Sun Ki Choi:** Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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